# H2 Math Summary

Notes

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# 1 Systems of Linear Equations

## 1.1 Forming the Equations

We read the context of the question and form equations depending on the context of the question.

**Example 1.1**

$$\left.\begin{aligned} a+b+c&=176 && —(1)\\ 0.85a+0.90b+0.95c&=161.80 && —(2)\\ 0.93a+0.95b+0.98c&=169.56 && —(3)\end{aligned}\right\} \Rightarrow a=32,\ b=44,\ c=100$$

**Example 1.2**

The "variables" may be expressions. To find $x$, $y$ and $z$, use the given information to form the equations:

$$\left.\begin{aligned} 2x^3+3\ln y-\mathrm{e}^z&=18 && —(1)\\ x^3-\ln y+2\mathrm{e}^z&=9 && —(2)\\ x^3+2\ln y+\mathrm{e}^z&=11 && —(3)\end{aligned}\right\} \Rightarrow x^3=8,\ \ln y=1,\ \mathrm{e}^z=1$$

It is technically still 'linear' because we can think of $A=x^3$, $B=\ln y$, $C=\mathrm{e}^z$, which turns it into a linear system in $A$, $B$ and $C$. A truly nonlinear equation could look like $x^2+x+y^2+2y-3z^3-xz+xy=1$; it cannot be made linear this way, because $x$ appears as both $x^2$ and $x$, and there are product terms such as $xz$ and $xy$.

## 1.2 Special Case: Number of Equations > Number of Variables

Sometimes the number of equations is more than the number of variables, and the solution must satisfy all of the equations.

**Example 1.3**

$$\left.\begin{aligned} \lambda+\mu&=1 && —(1)\\ 2\lambda-\mu&=5 && —(2)\end{aligned}\right\} \Rightarrow \lambda=2,\ \mu=-1$$

$$\lambda+3\mu=-1 \qquad —(3)$$

Sometimes there may be no solution, e.g. skew lines in vectors. In this case, solve for $\lambda$ and $\mu$ using equations (1) and (2). Then sub into (3).

> **Note**
>
> Check (3): if LHS = RHS, the equations are consistent; if LHS ≠ RHS, there is no solution.

## 1.3 Special Case: Number of Equations < Number of Variables

Sometimes the number of equations is less than the number of variables, so there are infinitely many solutions.

**Example 1.4**

$$\left.\begin{aligned} x+y+z=6 \quad &—(1) \\ x-y+2z=5 \quad &—(2) \end{aligned}\right\} \Rightarrow x=\tfrac{11}{2}-\tfrac{3}{2}z,\ y=\tfrac{1}{2}+\tfrac{1}{2}z,\ z=z$$

> **Note**
>
> Useful trick: in contextual questions, the variables are usually non-negative integers, $z \in \mathbb{Z}_0^+$, so use this to find the solution from the general solution.

When $z=z$, it means that there are infinitely many solutions, and $z$ can be any real value. So just sub that into $x$ and $y$ to get a correct solution.

$z=-1$: $(7,0,-1)$ $z=0$: $(\tfrac{11}{2},\tfrac{1}{2},0)$ $z=1$: $(4,1,1)$ $z=2$: $(\tfrac{5}{2},\tfrac{3}{2},2)$

$z=3$: $(1,2,3)$ $z=4$: $(-\tfrac{1}{2},\tfrac{5}{2},4)$ $z=5$: $(-2,3,5)$ $z=6$: $(-\tfrac{7}{2},\tfrac{7}{2},6)$

We can use the GC table to find all these solutions.

# 2 Inequalities

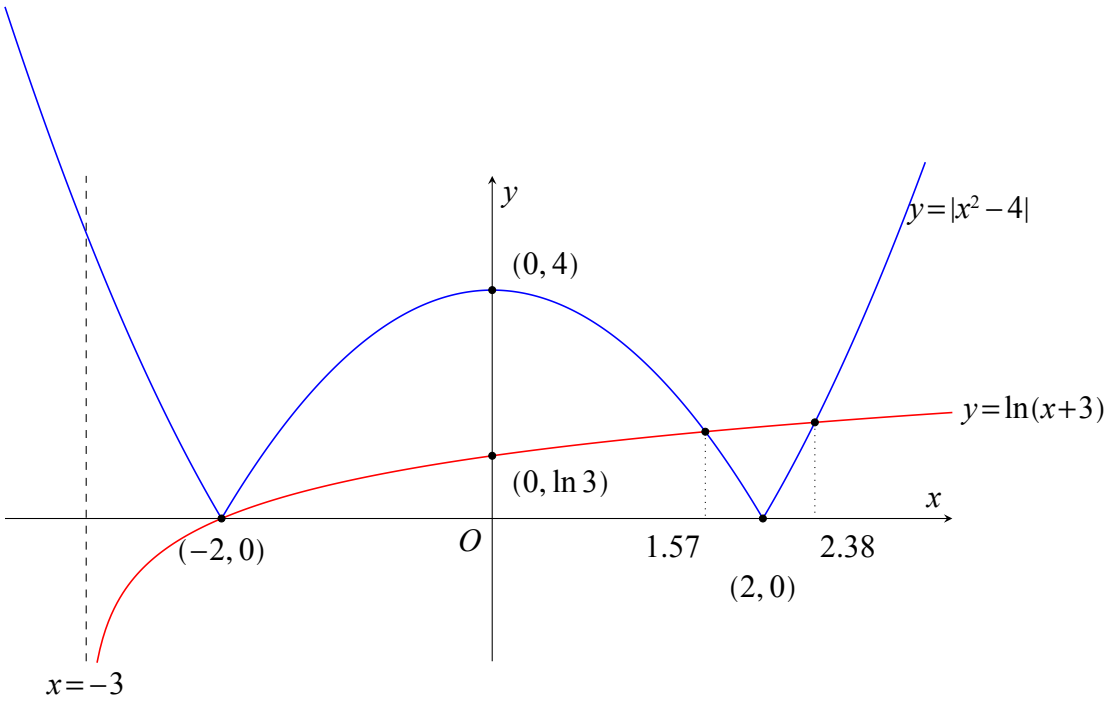
## 2.1 Solving Inequalities Graphically

Some inequalities cannot be solved algebraically, so we sketch the graphs and read off the solution.

- Rearrange to $f(x) > 0$, or compare $y = f(x)$ with $y = g(x)$.
- Sketch on the GC; mark the intersections and the asymptotes.
- Read off the $x$-values where the required graph is above or below.

**Example 2.1**

$$|x^2 - 4| \leqslant \ln(x+3)$$




Using GC, intersections at $x = 1.5746$, $x = 2.3840$

$$\therefore x = -2 \text{ or } 1.57 \leqslant x \leqslant 2.38$$

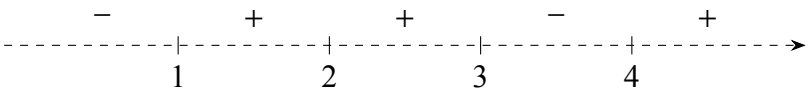
## 2.2 Sign Test Technique

We solve rational inequalities by finding where the expression changes sign.

- Make the right-hand side 0.
- Do **not** cross multiply unless the expression or bracket is always positive.
- Expressions that are always positive can be divided across, e.g. a quadratic with discriminant $<0$ and positive leading coefficient: $x^2+2x+5=(x+1)^2+4>0$.

**Example 2.2**

$$\frac{(x-1)(x-2)^2}{(x-3)(x-4)} \leqslant 0$$




$x=2$: numerator $=0$ (included)

$x=3,\ x=4$: denominator $=0$ (excluded)

$\therefore x \leqslant 1$ or $x=2$ or $3<x<4$

You can sketch the graph to check the answer.

> **Note**
>
> Crossing a factor with an odd power changes the sign; crossing a factor with an even power keeps the sign the same.

## 2.3 Common Substitutions

Once an inequality in $x$ is solved, we can replace $x$ with an expression to deduce the solution of a related inequality.

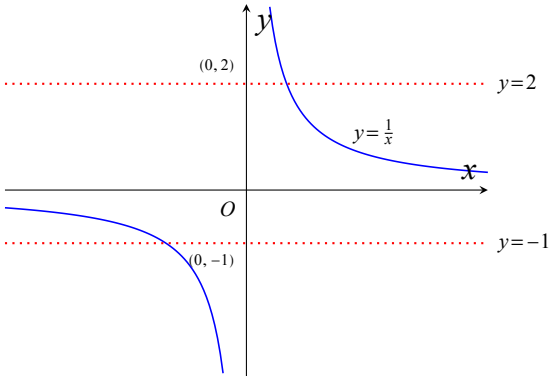
| **Replace $x$ by** | **Original inequality** | | |
|---|---|---|---|
| | $x>2$ | $x<-1$ | $-1<x<2$ |
| $e^x$ | $x>\ln 2$ | no solution | $x<\ln 2$ |
| $\ln x$ | $x>e^2$ | $0<x<e^{-1}$ | $e^{-1}<x<e^2$ |
| $-x$ | $x<-2$ | $x>1$ | $-2<x<1$ |
| $x+1$ | $x>1$ | $x<-2$ | $-2<x<1$ |
| $\lvert x\rvert$ | $x<-2$ or $x>2$ | no solution | $-2<x<2$ |
| $x^2$ | $x<-\sqrt{2}$ or $x>\sqrt{2}$ | no solution | $-\sqrt{2}<x<\sqrt{2}$ |
| $\frac{1}{x}$ | $\left(0<x<\frac{1}{2}\right)$ | $(-1<x<0)$ | $\left(x<-1 \text{ or } x>\frac{1}{2}\right)$ |
| $\cos x,\ 0\leqslant x\leqslant 2\pi$ | (no solution) | (no solution) | $(0\leqslant x<\pi \text{ or } \pi<x\leqslant 2\pi)$ |

Regardless of whatever substitution you use, using a graph will always work, but some can be solved without a graph.

For $\frac{1}{x}$, $\sin x$ and $\cos x$, draw the graph to solve.

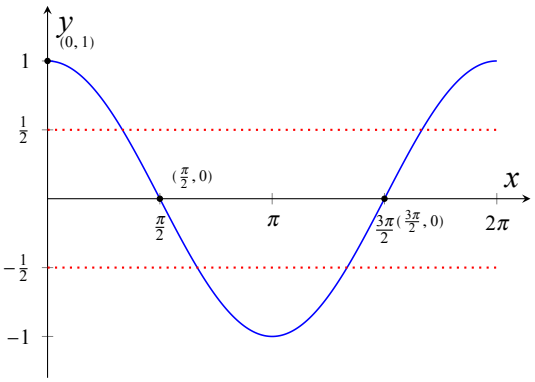
**Example 2.3**

$$\frac{1}{x}>2, \quad \frac{1}{x}<-1, \quad -1<\frac{1}{x}<2$$




**Example 2.4**

$$\cos x>\tfrac{1}{2}, \quad \cos x<-\tfrac{1}{2},$$
$$-\tfrac{1}{2}<\cos x<\tfrac{1}{2}, \quad 0\leqslant x\leqslant 2\pi$$

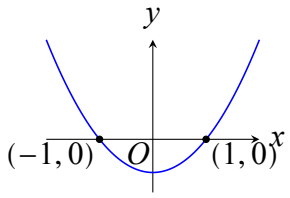


# 3 Graphing Techniques

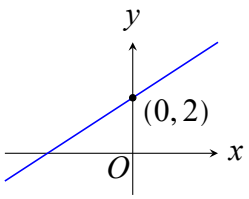
## 3.1 Key Features of a Curve

Key features of a curve:

(a) ***x*-intercepts:** let $y=0$.
e.g. $y=x^2-1$: $x^2-1=0 \Rightarrow x=\pm 1$.

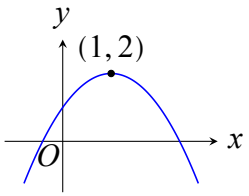



***y*-intercepts:** let $x=0$.
e.g. $y=x+2$: $y=2$.

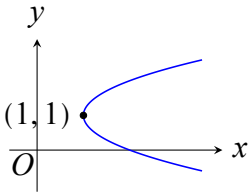



(b) **Stationary points**, when $\frac{dy}{dx}=0$ or $\frac{dy}{dx}$ is undefined. Label upon request.

**Horizontal turning point:** $\frac{dy}{dx}=0$ (tangent horizontal).
e.g. $y=-(x-1)^2+2$, maximum point $(1,2)$.

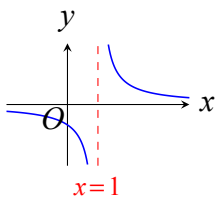
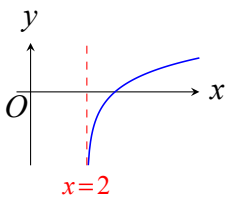
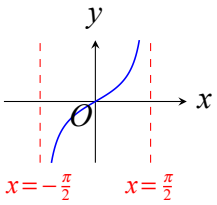



**Vertical turning point:** $\frac{dy}{dx}$ undefined (tangent vertical).
e.g. $x=(y-1)^2+1$, turning point $(1,1)$.



(c) **Asymptotes**

(i) **Vertical asymptotes.** If as $x \to k, y \to \pm\infty$, then $x = k$ is a vertical asymptote.

| **Common Occurrence** | **Condition** | **Example and asymptote** | **Graph** |
|---|---|---|---|
| Fractions | denominator $= 0$ | $y = \dfrac{1}{x-1}$<br>$x - 1 = 0 \Rightarrow x = 1$<br>$\therefore x = 1$ is a vertical asymptote |  |
| Logarithmic graphs | inside ln $= 0$ | $y = \ln(x-2)$<br>$x - 2 = 0 \Rightarrow x = 2$<br>$\therefore x = 2$ is a vertical asymptote |  |
| Tangent graphs | inside tan $= \dfrac{\pi}{2} + k\pi$,<br>$k \in \mathbb{Z}$ | $y = \tan x$<br>$x = \pm\dfrac{\pi}{2}, \pm\dfrac{3\pi}{2}, \ldots$ |  |

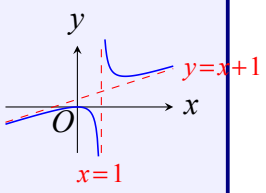
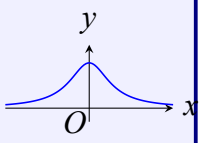
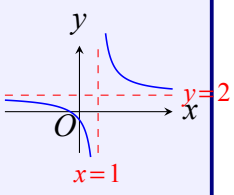
(ii) **Horizontal asymptotes.** As $x \to \pm\infty$, if $y \to k$, then $y = k$ is a horizontal asymptote.

- To find horizontal asymptotes, one should sub in $x \to \pm\infty$ into the equation. This is the only foolproof way to do it.
- A curve **CAN** intersect its horizontal asymptote.

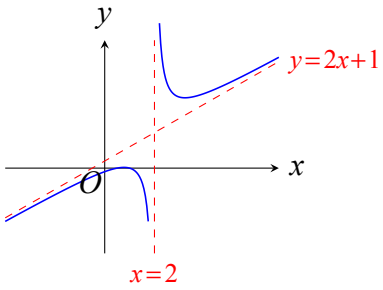
Note: there are more cases of horizontal asymptotes, but these are the common cases.

**Useful trick to find horizontal asymptotes**

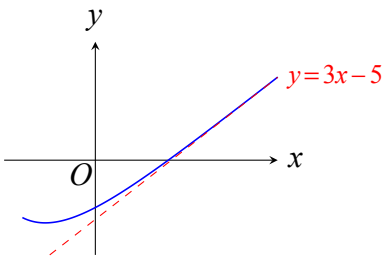
For rational functions $\frac{P(x)}{Q(x)}$:

| Case | Horizontal asymptote | Example | Graph |
|---|---|---|---|
| numerator power > denominator power | no horizontal asymptote | $y=\frac{x^2}{x-1}$ |   |
| numerator power < denominator power | $y=0$ | $y=\frac{1}{x^2+1}$ |   |
| numerator power = denominator power | ratio of coefficients of highest power | $y=\frac{2x+1}{x-1}$ $y=2$ |   |

(iii) **Oblique asymptotes.** An asymptote of the form $y=mx+c$, $m\neq 0$.




e.g. $y=2x+1+\frac{3}{x-2}$




e.g. $y=3x-5+e^{-x}$

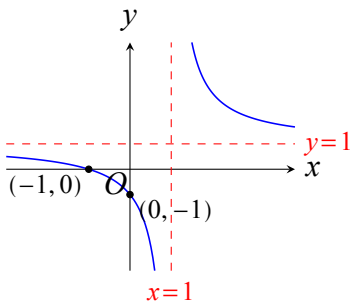
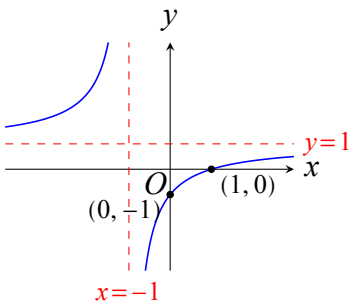
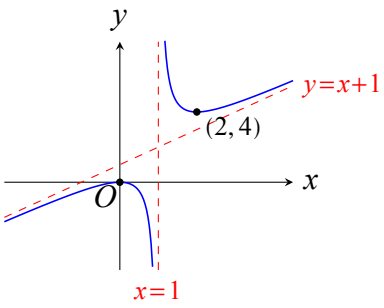
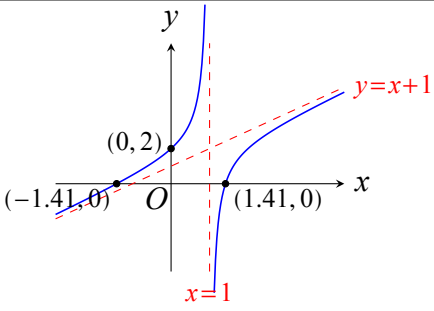
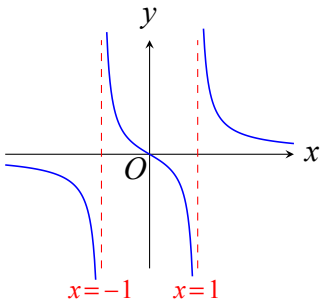
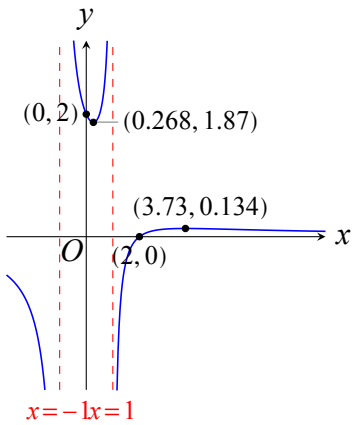
**Note**

For parametric curves, oblique asymptotes (if they exist) can only be found via the cartesian equation.

Graphs with oblique asymptotes come in the form $y=mx+c+g(x)$, where $g(x)\to 0$ as $x\to\pm\infty$; the oblique asymptote is $y=mx+c$.

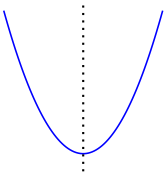
## 3.2 Graphs of Rational Functions

Two common shapes for each type.

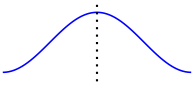
| Type and form | Shape 1 | Shape 2 |
|---|---|---|
| **Linear / linear**<br>$y=\dfrac{Ax+B}{Cx+D}$<br>Horizontal asymptote: $y=\dfrac{A}{C}$<br>Vertical asymptote: $x=-\dfrac{D}{C}$ |  <br>$y=\dfrac{x+1}{x-1}$ |  <br>$y=\dfrac{x-1}{x+1}$ |
| **Quadratic / linear**<br>$y=\dfrac{ax^2+bx+c}{dx+e}$<br>$=Ax+B+\dfrac{C}{dx+e}$<br>Oblique asymptote: $y=Ax+B$<br>Vertical asymptote: $x=-\dfrac{e}{d}$ |  <br>2 turning points<br>$y=x+1+\dfrac{1}{x-1}\quad\left(\dfrac{Cd}{A}>0\right)$ |  <br>No turning points<br>$y=x+1-\dfrac{1}{x-1}\quad\left(\dfrac{Cd}{A}<0\right)$ |
| **Linear / quadratic**<br>$y=\dfrac{ax+b}{cx^2+dx+e}$<br>Horizontal asymptote: $y=0$<br>Vertical asymptotes: solve $cx^2+dx+e=0$<br>(possibly no roots) |  <br>$y=\dfrac{x}{(x-1)(x+1)}$ |  <br>$y=\dfrac{x-2}{(x-1)(x+1)}$ |

## 3.3 Symmetry

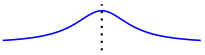
**Even functions:** $\mathrm{f}(-x)=\mathrm{f}(x)$. The graph is symmetric about the $y$-axis (line of symmetry $x=0$, dotted).



e.g. $y=x^2$



e.g. $y=\cos x$

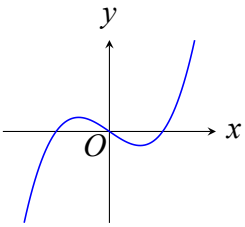


e.g. $y=\dfrac{1}{x^2+1}$

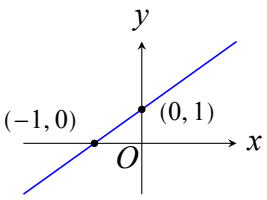
**Odd functions:** $\mathrm{f}(-x)=-\mathrm{f}(x)$.

The graph is symmetric about the origin.

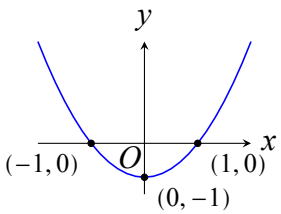
e.g. $y=x^3-x$



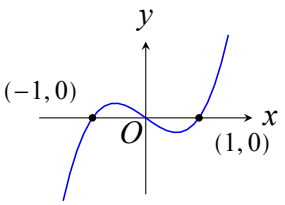
## 3.4 Basic Graphs



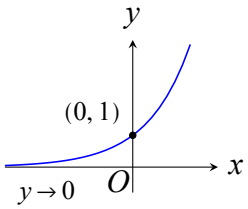

Linear $y=x+1$



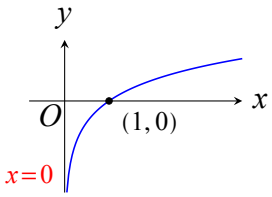

Quadratic $y=x^2-1$



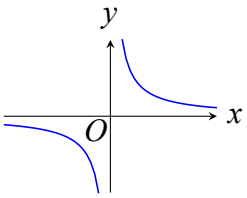

Cubic $y=x^3-x$



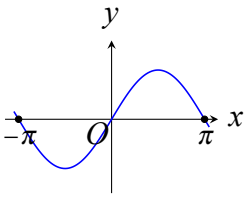

Exponential $y=e^x$



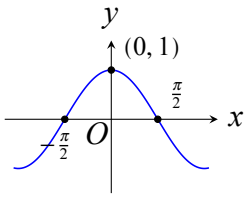

Logarithm $y=\ln x$



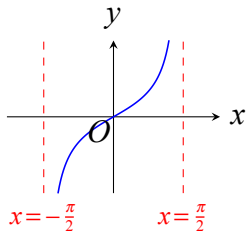

Reciprocal $y=\frac{1}{x}$



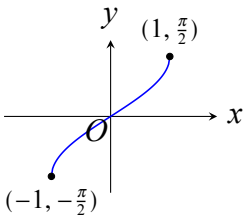

$y=\sin x$



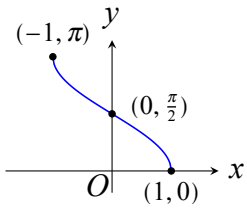

$y=\cos x$



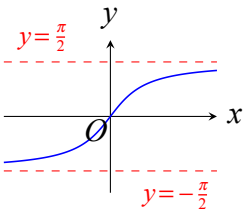

$y=\tan x$




$y=\sin^{-1}x$




$y=\cos^{-1}x$




$y=\tan^{-1}x$

## 3.5 Parametric Equations

A curve $C$ can be described by giving both coordinates in terms of a parameter $t$:

$$x=x(t), \quad y=y(t), \quad t\in T$$

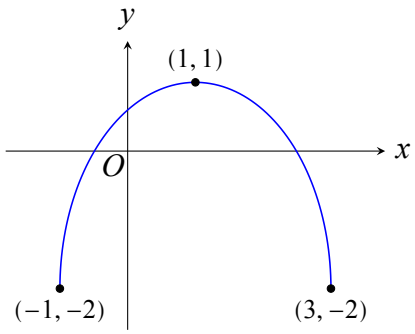
**Example.**

$x=2\cos t+1, \quad y=3\sin t-2, \quad 0\leqslant t\leqslant \pi$

$t=0$: $(3,-2)$ $\quad t=\frac{\pi}{2}$: $(1,1)$ $\quad t=\pi$: $(-1,-2)$

$x=0$: $\left(0, \frac{3\sqrt{3}}{2}-2\right)$

$y=0$: $\left(1\pm\frac{2\sqrt{5}}{3}, 0\right)$




**CAUTION: sketching from parametric equations**

**ALWAYS take note of the range of $t$ when sketching graphs.** The restriction on $t$ decides which part of the curve exists: above, $0\leqslant t\leqslant\pi$ gives only the upper half of the ellipse, with endpoints $(3,-2)$ and $(-1,-2)$, not the whole ellipse.

### 3.5.1 Finding the Cartesian Equation

The cartesian equation of $C$ is an equation without the parameter $t$. Common techniques:

① **Make $t$ the subject.** e.g. $x=2t-1, y=3t^2+e^t, t\in\mathbb{R}$

$$t=\frac{x+1}{2}$$

$$y=3\left(\frac{x+1}{2}\right)^2+e^{\frac{x+1}{2}}$$

② **Trigo identity formulas.** e.g. $x=2\cos t+1, y=3\sin t-2, 0\leqslant t\leqslant\pi$

$$\cos t=\frac{x-1}{2}, \quad \sin t=\frac{y+2}{3}$$

$$\cos^2 t+\sin^2 t=1$$

$$\frac{(x-1)^2}{4}+\frac{(y+2)^2}{9}=1, \quad y\geqslant -2$$

③ **Others.** e.g. $x=t-\frac{1}{t}, y=t+\frac{1}{t}, t\neq 0$

$$y+x=2t, \quad y-x=\frac{2}{t}$$

$$(y+x)(y-x)=4$$

$$y^2-x^2=4$$

Note: Avoid finding the cartesian eqn if the question did not ask you to.

### 3.5.2 Asymptotes

- Find (if possible) $t \to t_0$ such that $x \to x_0, y \to \pm\infty$ (vertical asymptote $x = x_0$), or $y \to y_0, x \to \pm\infty$ (horizontal asymptote $y = y_0$).
- Oblique asymptotes (if they exist) can only be found via the cartesian equation.

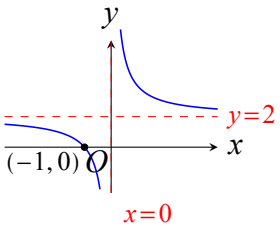
**Example.** $x = \dfrac{1}{t-1},\ y = 2t,\ t \neq 1$

As $t \to \pm\infty,\ x \to 0,\ y \to \pm\infty$

$\therefore x = 0$ is a vertical asymptote

As $t \to 1,\ x \to \pm\infty,\ y \to 2$

$\therefore y = 2$ is a horizontal asymptote



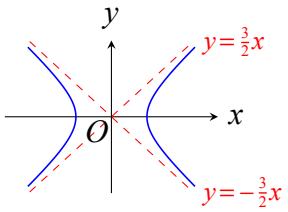

**Oblique asymptotes.** Cannot be found from the parametric form alone; find the cartesian equation first.

e.g. $x = 2\sec\theta,\ y = 3\tan\theta$

$$\sec^2\theta - \tan^2\theta = 1$$

$$\frac{x^2}{4} - \frac{y^2}{9} = 1$$

Oblique asymptotes: $y = \pm\dfrac{3}{2}x$




### 3.5.3 Intersections

Intersection between a parametric curve and another curve: sometimes both equations are needed.

e.g. $x = 2t + \ln t,\ y = t^2 - 1,\ t > 0$ and the circle $x^2 + y^2 = 25$:

$$(2t + \ln t)^2 + (t^2 - 1)^2 = 25$$

Using GC, $t = 0.00735$ or $t = 1.86$ (both $> 0$)

$\therefore$ intersections: $(-4.90,\ -1.00)$ and $(4.35,\ 2.47)$

This equation in $t$ cannot be solved algebraically, so the GC is needed.

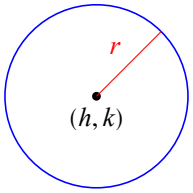
# 3.6 Conics

## Circle

$$(x-h)^2+(y-k)^2=r^2$$

Centre: $(h,k)$

Radius: $r$



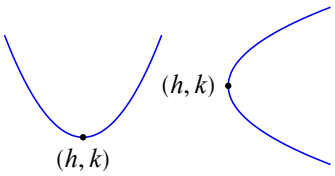

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## Parabola

$$y-k=a(x-h)^2 \qquad x-h=b(y-k)^2$$

$y-k=a(x-h)^2$:
turning point $(h,k)$, $\dfrac{dy}{dx}=0$

$x-h=b(y-k)^2$:
turning point $(h,k)$, $\dfrac{dy}{dx}$ undefined




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## Ellipse

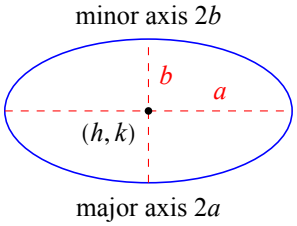
$$\frac{(x-h)^2}{a^2}+\frac{(y-k)^2}{b^2}=1$$

Centre: $(h,k)$

"Horizontal radius" $=a$, "vertical radius" $=b$

For $a>b$: major axis horizontal (length $2a$, semi-major $a$); minor axis vertical (length $2b$, semi-minor $b$).

⋆ RHS ⇒ numerator coefficients of $x^2$, $y^2$ must all be 1.




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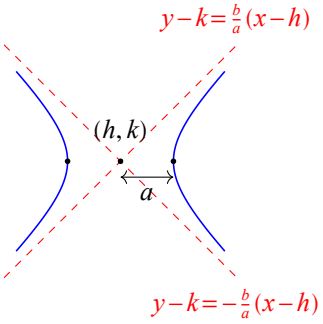
### Hyperbola: Type I

$$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$$

Centre: $(h, k)$

Distance from centre to vertices $= a$

Asymptotes (dashed): $y - k = \pm\frac{b}{a}(x-h)$




- RHS ⇒ numerator coefficients of $x^2$, $y^2$ must all be 1.
- Equation of asymptotes: $y - k = \pm\frac{b}{a}(x-h)$

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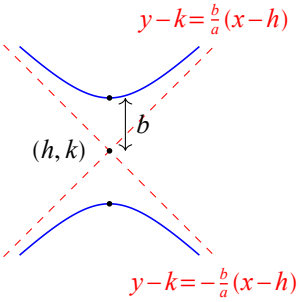
### Hyperbola: Type II

$$\frac{(y-k)^2}{b^2} - \frac{(x-h)^2}{a^2} = 1$$

Centre: $(h, k)$

Distance from centre to vertices $= b$

Asymptotes (dashed): $y - k = \pm\frac{b}{a}(x-h)$




- RHS ⇒ numerator coefficients of $x^2$, $y^2$ must all be 1.
- Equation of asymptotes: $y - k = \pm\frac{b}{a}(x-h)$

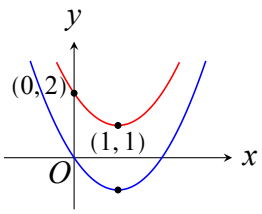
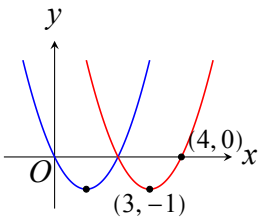
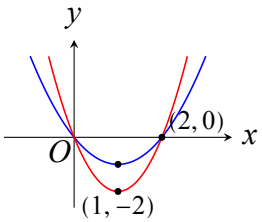
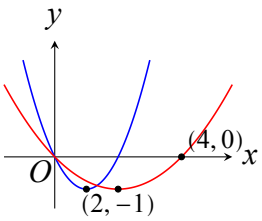
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# 4 Graphing Transformations

Every transformation comes from **replacing** $x$ **or** $y$ in the equation $y=\mathrm{f}(x)$.

Throughout, blue is $y=\mathrm{f}(x)=x(x-2)$, with intercepts $(0,0)$, $(2,0)$ and minimum point $(1,-1)$; red is the transformed graph.

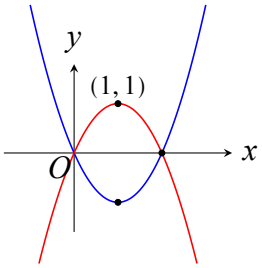
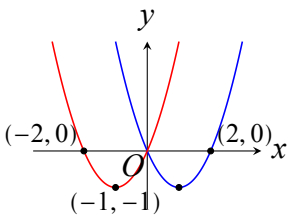
## 4.1 Translations and Scalings

| Transformation | Replacement | Result | Graph |
|---|---|---|---|
| Translation of $a$ units in the positive $y$ direction | replace $y$ with $y-a$: $y-a=\mathrm{f}(x)$ | $y=\mathrm{f}(x)+a$ (graph: $a=2$) |   |
| Translation of $a$ units in the positive $x$ direction | replace $x$ with $x-a$ | $y=\mathrm{f}(x-a)$ (graph: $a=2$) |   |
| Scaling by factor $a$ parallel to the $y$-axis | replace $y$ with $\frac{y}{a}$: $\frac{y}{a}=\mathrm{f}(x)$ | $y=a\mathrm{f}(x)$ (graph: $a=2$) |   |
| Scaling by factor $a$ parallel to the $x$-axis | replace $x$ with $\frac{x}{a}$ | $y=\mathrm{f}\left(\frac{x}{a}\right)$ (graph: $a=2$) |   |

**CAUTION**

Every replacement acts the "opposite" way to what it looks like: $x-2$ moves the graph in the **positive** $x$ direction; $y=\mathrm{f}(2x)$ is a scaling by factor $\frac{1}{2}$ (not 2) parallel to the $x$-axis.
Replacing $x$ does **not** mean replacing the input of f, and replacing $y$ does not mean replacing the output $\mathrm{f}(x)$.

## 4.2 Reflections

| Transformation | Replacement | Result | Graph |
|---|---|---|---|
| Reflection about the $x$-axis | replace $y$ with $-y$: $-y=\mathrm{f}(x)$ | $y=-\mathrm{f}(x)$ |   |
| Reflection about the $y$-axis | replace $x$ with $-x$ | $y=\mathrm{f}(-x)$ |   |

## 4.3 Combining Transformations

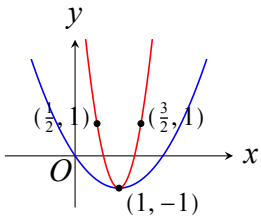
Functions of the form $y=a\mathrm{f}(bx+c)+d,\ \ a,b,c,d\in\mathbb{R}$: replace $x$ and $y$ **one step at a time**.

e.g. $y=\mathrm{f}(x)\ \longrightarrow\ y=2\mathrm{f}(2x-1)+1$

| Replacement | Transformation | Function |
|---|---|---|
| replace $x$ with $x-1$ | translation of 1 unit in the positive $x$ direction | $y=\mathrm{f}(x-1)$ |
| replace $x$ with $2x$ | scaling by factor $\frac{1}{2}$ parallel to the $x$-axis | $y=\mathrm{f}(2x-1)$ |
| replace $y$ with $\frac{y}{2}$ | scaling by factor 2 parallel to the $y$-axis | $y=2\mathrm{f}(2x-1)$ |
| replace $y$ with $y-1$ | translation of 1 unit in the positive $y$ direction | $y=2\mathrm{f}(2x-1)+1$ |

For $f(x)=x(x-2)$:

$$(1,-1) \to (2,-1) \to (1,-1)$$
$$\to (1,-2) \to (1,-1)$$
$$(0,0) \to (1,0) \to (\tfrac{1}{2},0)$$
$$\to (\tfrac{1}{2},0) \to (\tfrac{1}{2},1)$$

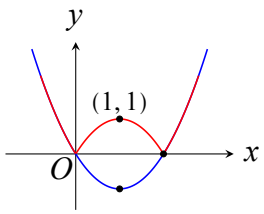
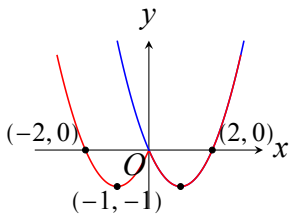



**CAUTION: order of transformations**

Inside f: do the translation **before** the scaling/reflection in $x$. $y=f(2x-1)$ is "translate 1 unit, then scale by $\frac{1}{2}$", **not** "scale by $\frac{1}{2}$, then translate 1 unit" (that gives $y=f(2(x-1))=f(2x-2)$).
Outside f: do the scaling before the translation in $y$.

## 4.4 Modulus Graphs

Every modulus graph can be partitioned into a "+" part and a "−" part: $|x|=x$ for $x \geqslant 0$, $|x|=-x$ for $x<0$.

| Graph | How to sketch | Graph |
|---|---|---|
| $y=\|f(x)\|$ | Keep the part on or above the $x$-axis; reflect the part below the $x$-axis about the $x$-axis.<br>Sharp points at the $x$-intercepts. |   |
| $y=f(\|x\|)$ | Remove the part on the left of the $y$-axis; reflect the part on the right of the $y$-axis about the $y$-axis. |   |

e.g. $y=f(x) \longrightarrow y=|f(-2x+1)|$

| Replacement | Transformation | Function |
|---|---|---|
| replace $x$ with $x+1$ | translation of 1 unit in the negative $x$ direction | $y=f(x+1)$ |
| replace $x$ with $-x$ | reflection about the $y$-axis | $y=f(-x+1)$ |
| replace $x$ with $2x$ | scaling by factor $\frac{1}{2}$ parallel to the $x$-axis | $y=f(-2x+1)$ |
| $\|f\|$ | reflect parts below the $x$-axis about the $x$-axis | $y=\|f(-2x+1)\|$ |

**CAUTION**

$y=\mathrm{f}(|x|+1)$ and $y=\mathrm{f}(|x+1|)$ are **different** curves.

$\mathrm{f}(|x|+1)$: translate first, then apply $|x|$. $\mathrm{f}(|x+1|)$: apply $|x|$ first, then translate.

## 4.5 Reciprocal Graphs $y=\frac{1}{\mathrm{f}(x)}$

Analyse the new curve at the critical points of $y=\mathrm{f}(x)$.

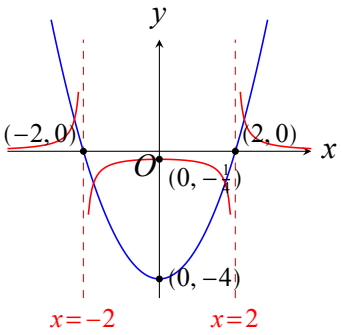
| $y=\mathrm{f}(x)$ | $y=\frac{1}{\mathrm{f}(x)}$ |
|---|---|
| $x$-intercept $(a,0)$ | vertical asymptote $x=a$ |
| vertical asymptote $x=a$ | $y\to 0$: point $(a,0)$ (excluded) |
| maximum point $(p,q)$, $q\neq 0$ | minimum point $\left(p,\frac{1}{q}\right)$ |
| minimum point $(p,q)$, $q\neq 0$ | maximum point $\left(p,\frac{1}{q}\right)$ |
| horizontal asymptote $y=k$, $k\neq 0$ | horizontal asymptote $y=\frac{1}{k}$ |
| $y\to\pm\infty$ (incl. oblique asymptote) | horizontal asymptote $y=0$ |
| $y=\pm 1$ | same point $(\frac{1}{\pm 1}=\pm 1)$ |
| $\mathrm{f}(x)>0$ / $\mathrm{f}(x)<0$ | $\frac{1}{\mathrm{f}(x)}>0$ / $\frac{1}{\mathrm{f}(x)}<0$ (same sign) |

e.g. $\mathrm{f}(x)=x^2-4$:

$(-2,0),\ (2,0)\to x=\pm 2$

$(0,-4)\to\left(0,-\frac{1}{4}\right)$

$y\to\infty\to y=0$




## 4.6 Derivative Graphs $y=\mathrm{f}'(x)$

$\mathrm{f}'(x)=\frac{\mathrm{d}y}{\mathrm{d}x}$ is the gradient of $y=\mathrm{f}(x)$: read off the gradient on each interval.

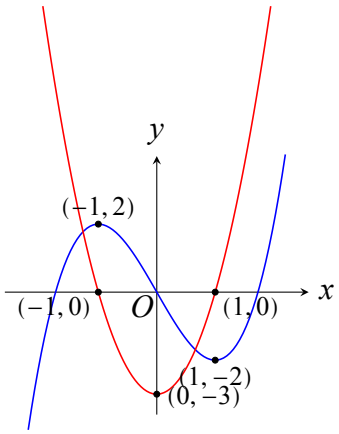
| $y=\mathrm{f}(x)$ | $y=\mathrm{f}'(x)$ |
|---|---|
| stationary point at $x=a$ | $x$-intercept $(a,0)$ |
| increasing / decreasing | above / below the $x$-axis |
| point of inflexion (max. or min. gradient) | turning point |
| vertical asymptote $x=a$ | vertical asymptote $x=a$ |
| horizontal asymptote $y=k$ | horizontal asymptote $y=0$ |
| oblique asymptote $y=mx+c$ | horizontal asymptote $y=m$ |

e.g. $\mathrm{f}(x)=x^3-3x$:

$$\mathrm{f}'(x)=3x^2-3$$

SP at $x=\pm 1 \rightarrow (\pm 1, 0)$

inflexion at $x=0 \rightarrow$ min. $(0,-3)$




**CAUTION**

Mark $x$-coordinates only where they come from $y=\mathrm{f}(x)$: the $y$-values of $y=\mathrm{f}'(x)$ are generally **not** known from the graph of $y=\mathrm{f}(x)$.

## 4.7 Reverse Transformations

If $y=f(x)$ undergoes a sequence of transformations to give a known $y=g(x)$, undo the transformations **in reverse order**, each with its opposite replacement.

e.g. translate 2 units in the negative $x$ direction, reflect about the $y$-axis, scale by factor 3 parallel to the $y$-axis; result $y=6e^{-x+3}$.

| **Opposite transformation** | **Replacement** | **Function** |
|---|---|---|
| scale by factor $\frac{1}{3}$ parallel to the $y$-axis | replace $y$ with $3y$ | $y=2e^{-x+3}$ |
| reflect about the $y$-axis | replace $x$ with $-x$ | $y=2e^{x+3}$ |
| translate 2 units in the positive $x$ direction | replace $x$ with $x-2$ | $y=2e^{x+1}$ |

$$\therefore f(x)=2e^{x+1}$$

## 4.8 Transformations of Conics

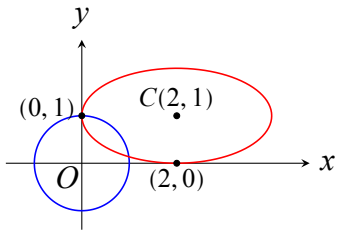
Apply exactly the same replacements of $x$ and $y$.

e.g. $x^2+y^2=1$: scale by factor 2 parallel to the $x$-axis, translate 2 units in the positive $x$ direction, translate 1 unit in the positive $y$ direction.

| **Transformation** | **Replacement** | **Equation** |
|---|---|---|
| scale by factor 2 parallel to the $x$-axis | replace $x$ with $\frac{x}{2}$ | $\frac{x^2}{4}+y^2=1$ |
| translate 2 units in the positive $x$ direction | replace $x$ with $x-2$ | $\frac{(x-2)^2}{4}+y^2=1$ |
| translate 1 unit in the positive $y$ direction | replace $y$ with $y-1$ | $\frac{(x-2)^2}{4}+(y-1)^2=1$ |

$$\therefore \frac{(x-2)^2}{4}+(y-1)^2=1$$

Centre $(2, 1)$.




**CAUTION**

Replace $x$ by $(x-2)$ **inside** the bracket: $\frac{(x-2)^2}{4}$, not $\frac{x^2}{4}-2$ or $\frac{x^2-2}{4}$.